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Dear Editor,

We are pleased to submit our manuscript entitled:

"Topology-Aware GPCR–G Protein Coupling Prediction via ESM-2 Embeddings and Curated Transmembrane Annotations"

for consideration as a **Method Article** in *Briefings in Bioinformatics.*

Summary of Contribution G protein-coupled receptors (GPCRs) are the largest family of membrane proteins and the target of approximately 30% of FDA-approved drugs. Predicting which G protein a GPCR will couple to is a long-standing problem in computational biology. Existing sequence-based methods often treat the task as single-protein classification and rely on flat embeddings from protein language models, which can misallocate attention to non-binding regions such as the N- and C-terminal tails.

In this study, we make three advances:

1. **Genuine pairwise prediction at scale.** We constructed a dataset of 1,639 (GPCR, G-protein-family) pairs spanning 448 non-redundant receptors and four families (Gq, Gi, Gs, G12/13). By allowing the same GPCR to appear with different G proteins, we eliminated the degeneracy of fixed-template prediction and ensured that both receptor and G-protein embeddings contribute to the decision boundary.
2. **Topology-aware feature engineering.** We fetched curated trans-

membrane annotations from UniProt (98.8% 7-TM coverage) to precisely delimit ICL2 and ICL3—the cytoplasmic loops known to mediate G-protein binding. Injecting physicochemical statistics and local ESM embeddings of these loops improved homology-aware cluster-based CV AUC from **0.797 to 0.830** and redirected model attention from the C-tail (20% of SHAP-mapped residues in the baseline) toward the true interface.

3. **Rigorous evaluation and interpretability.** We employed cluster-aware cross-validation (guaranteeing unseen GPCRs in each fold) and leave-one-G-protein-family-out (LOGPSO) tests. Permutation-importance analysis showed that ICL3 length and ICL2 aromatic ratio ranked higher than the average global ESM-2 dimension, offering a low-dimensional, biologically grounded augmentation strategy for membrane-protein prediction.

Why Briefings in Bioinformatics? We believe this work is particularly suitable for *Briefings in Bioinformatics* for the following reasons:

- **Methodological rigor:** The combination of large-scale pairing data, homology-aware splitting, and cross-family validation addresses a critical gap in how GPCR–G protein coupling benchmarks are constructed.
- **Novel interpretability angle:** We demonstrate that even powerful pre-trained embeddings (ESM-2) can benefit from explicit, curator-grade topological guidance—a finding transferable to other membrane-protein prediction tasks.
- **Practical utility:** The 16-dimensional ICL statistics alone contribute approximately 39% of the cluster-aware gain, with the remaining 61% coming from local ESM embeddings within the same loops. Together, the full topology-aware feature set improves AUC from 0.797 to 0.830, providing a lightweight, experimentally testable augmentation strategy.
- **Reproducibility:** All code, pairing matrices, cluster assignments, and evaluation scripts are made publicly available.

Novelty and Impact While previous studies have applied deep learning to GPCR–G protein coupling, our work is the first to:

- Frame the task as genuine multi-family pairwise prediction at this scale (448 receptors);

- Use UniProt-curated transmembrane annotations to extract precise ICL2/3 features and quantify their impact under strict homology-aware validation;
- Show that topology correction redirects pre-trained model attention from non-binding tails to the known G-protein interface, as validated by SHAP and permutation importance.

Related Publications This manuscript represents original work that has not been published or submitted elsewhere. All authors have approved the manuscript and agree with its submission to *Briefings in Bioinformatics*.

Conflicts of Interest The authors declare no competing financial interests.

Thank you for considering our submission. We look forward to your response.

Qingyong Wang and Lichuan Gu

Sincerely,

Qingyong Wang and Lichuan Gu

On behalf of all co-authors